ECO 362: Financial Investments
Risk and Return

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Risk and return

- Financial investments are fundamentally about trading off risk with expected returns.
- Consider two assets.
  1. 1-month T-bill, which have monthly returns $R_{b,t}$.
  2. A value-weighted portfolio of all US stocks, which have monthly returns

$$R_{s,t} = \sum_i w_{i,t-1} R_{i,t}$$

     where $w_{i,t-1}$ is stock i's market cap at month $t-1$ as a share of total market cap.

Vanguard Total Stock Market Index Fund ETF (VTI)

Holdings

Portfolio Holdings	No.	%
Equity Holding	4,052	99.83
Bond Holding	0	0.00
Other Holding	7	0.17

% Assets in Top 10 Holdings
23

Reported Turnover %
4.00
As of 12/31/21

Women Directors %
32

Women Executives %
26

Holdings	% Portfolio Weight	First Bought	Market Value USD as of Aug 31, 2022	1-Year Return	Forward P/E	Equity Star Rating	Economic Moat	ESG Risk Rating Assessment	Sector
Apple Inc	6.10	Mar 31, 2004	71,719,148,964	0.39	22.57	★★★	Narrow		Technology
Microsoft Corp	4.94	Mar 31, 2004	58,017,307,679	-15.50	23.53	★★★★	Wide		Technology
Amazon.com Inc	2.77	Dec 31, 2003	32,528,451,044	-30.45	47.62	★★★★	Wide		Consumer Cyclical
Tesla Inc	1.84	Dec 31, 2010	21,603,533,855	2.98	45.45	★★★	Narrow		Consumer Cyclical
Alphabet Inc Class A	1.65	Sep 30, 2004	19,407,107,210	-27.49	16.13	★★★★	Wide		Communication Services
Alphabet Inc Class C	1.46	May 31, 2018	17,160,806,514	-27.08	16.23	★★★★	Wide		Communication Services
UnitedHealth Group Inc	1.23	Dec 31, 2003	14,451,849,285	28.63	20.49	★★	Narrow		Healthcare
Berkshire Hathaway Inc Class B	1.19	Mar 31, 2010	13,950,689,033	-2.78	18.59	★★★★★	Wide		Financial Services
Johnson & Johnson	1.07	Dec 31, 2003	12,593,573,755	2.98	15.55	★★★	Wide		Healthcare
Exxon Mobil Corp	1.02	Dec 31, 2003	11,943,667,958	53.97	8.26	★★★	Narrow		Energy

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Holdings as of Aug 31, 2022 | The top 100 largest holdings are available for display across Equity, Bond and Other.

Source: Morningstar Direct

A statistics refresher

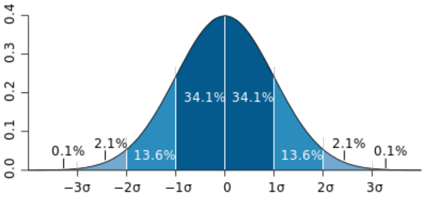
- Mean of stock returns is

$$\mu = \frac{1}{T}\sum_{t=1}^{T} R_{s,t}$$

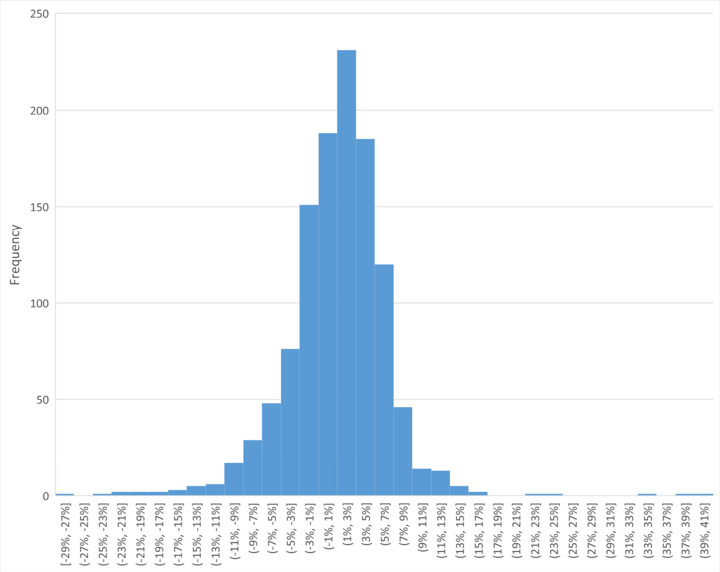
- Variance of stock returns is

$$\sigma^2 = \frac{1}{T}\sum_{t=1}^{T}(R_{s,t} - \mu)^2$$

- Standard deviation σ is the square root of variance.



Monthly stock returns (1926–2022)



Example 1: Two strategies

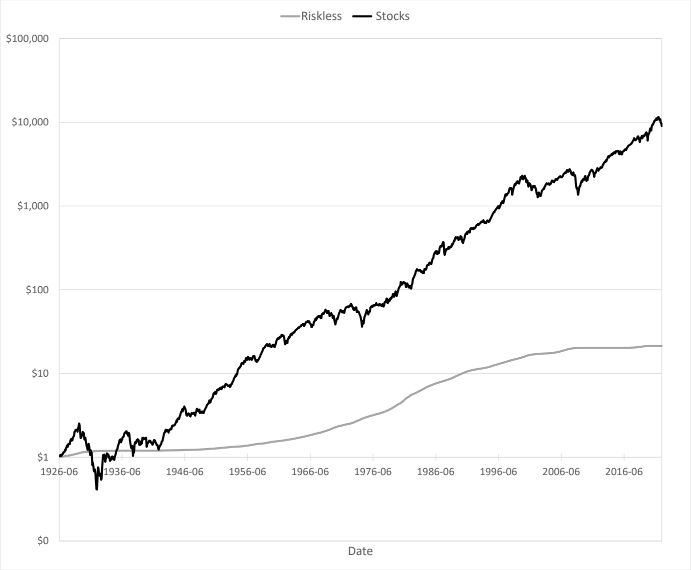
- Start with \$1 in June 1926.
- How would your wealth have grown through June 2022 under two strategies.
  1. Invest in 1-month T-bills. The cumulative return is

$$\prod_{t=1926:07}^{2022:06} (1 + R_{b,t})$$

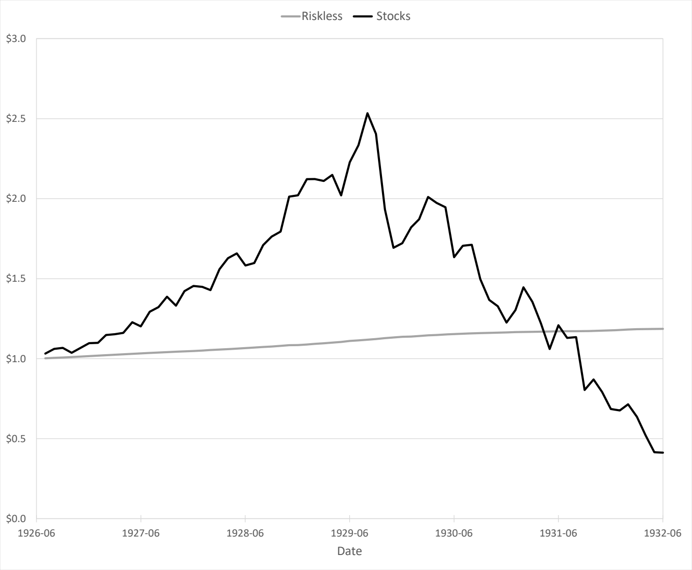
  2. Invest in the US stock portfolio. The cumulative return is

$$\prod_{t=1926:07}^{2022:06} (1 + R_{s,t})$$

Cumulative return (1926–2022)



Cumulative return (1926–1932)



Which is a better strategy?

- You end up with more wealth in the long run if you invest in the US stock market (\$9,028) rather than T-bills (\$21.38).
- However, the path that you took is more uncertain.

Statistic	T-bills	Stocks
Mean	0.27%	0.94%
Standard deviation	0.25%	5.33%

Portfolio

- You do not have to put your wealth exclusively in T-bills or stocks.
- Why not create a portfolio?

$$\begin{aligned} R_p =& (1-w)R_b + wR_s \\ =& R_b + w(R_s - R_b) \end{aligned}$$

- At the end of June 2022, your job is to choose w.
- Monthly yield on the 1-month T-bill in June 2022 is 0.08%.
  - Thus, $R_b = 0.08\%$ in July 2022 with certainty (i.e., standard deviation of 0).
- Stock return R_s in July 2022 is random.
  - Historically, stocks have $\mu = 0.94\%$ and $\sigma = 5.33\%$.
  - Assume that this is a reliable forecast of what to expect in July 2022.

Tradeoff between risk and return

- The expected portfolio return is

$$\begin{aligned}\mathbb{E}[R_p] =& R_b + w(\mu - R_b)\\ =& 0.08\% + w(0.94\% - 0.08\%)\end{aligned}$$

- The variance is

$$\begin{aligned}\mathrm{Var}(R_p) =& \mathrm{Var}(R_b + w(R_s - R_b)) = w^2\sigma^2\\ =& w^2(5.33\%)^2\end{aligned}$$

- The standard deviation is

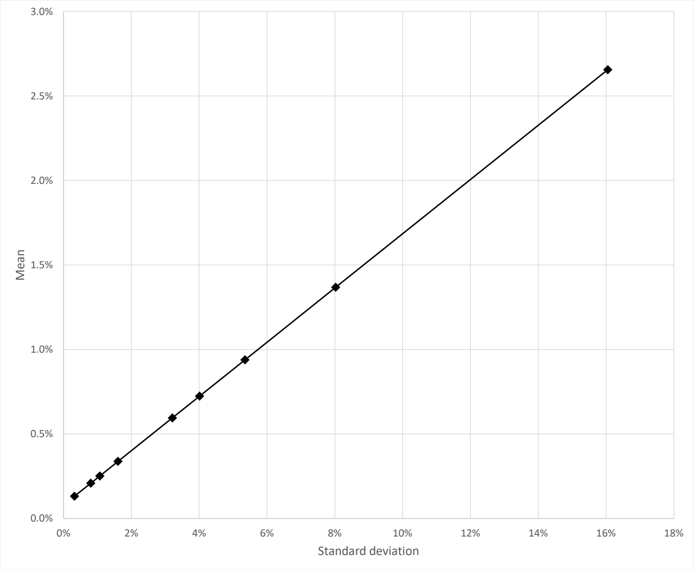
$$\sigma(R_p) = w\sigma = w \times 5.33\%$$

- Combine the equations for mean and standard deviation by substituting out w.

$$\begin{aligned}\mathbb{E}[R_p] =& R_b + \underbrace{\frac{\mu - R_b}{\sigma}}_{\text{Sharpe ratio}} \sigma(R_p) \\ =& 0.08\% + \frac{0.94\% - 0.08\%}{5.33\%}\sigma(R_p)\end{aligned}$$

- This is a linear line (recall $y = a + bx$).
- The intercept is the riskless rate.
- The slope is called the Sharpe ratio.
- The higher is the Sharpe ratio, the better is the tradeoff (i.e., higher mean per unit of risk).

Tradeoff between risk and return



What is the optimal portfolio?

- Equivalent to asking which point on the line would you choose?
- Investors choose a different w depending on risk aversion.
  - More risk tolerant: Go for higher mean and variance by choosing a higher w.
  - More risk averse: Go for lower mean and variance by choosing a lower w.
- In economics, expected utility theory ranks preferences for risk vs. return.

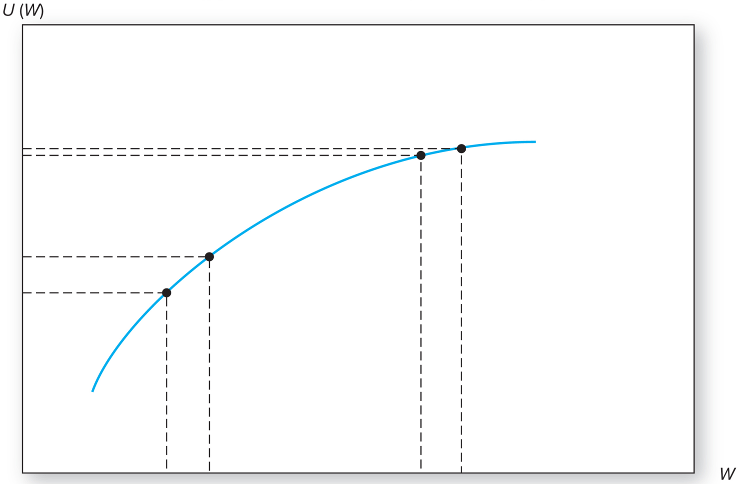
Portfolio-choice problem

- Start with 1 unit (e.g., \$1k, \$1m, or \$1b) of wealth.
- Maximize expected utility of wealth after a period (e.g., 1 month, 1 year, or 10 years).

$$\max_{w} \ \mathbb{E}[U(1 + R_p)]$$

  where $R_p = R_b + w(R_s - R_b)$.
- The answer depends on the utility function $U(\cdot)$.
  - Increasing ($U' > 0$): More wealth is better.
  - Concave ($U'' < 0$): Marginal utility of wealth decreases as you get richer.

Illustration of diminishing marginal utility



Source: Bodie, Kane, and Marcus (2018, Figure 6.1)

Mean-variance expected utility

- Quadratic utility:

$$U(1 + R_p) = 1 + R_p - \frac{\gamma}{2}(R_p - \mathbb{E}[R_p])^2$$

  where $\gamma \geq 0$.

- Mean-variance expected utility:

$$\mathbb{E}[U(1 + R_p)] = 1 + \mathbb{E}[R_p] - \frac{\gamma}{2}\mathrm{Var}(R_p)$$

What is your risk aversion?

- The annual riskless rate is 1%.
- The annual standard deviation of stock returns is 18%.
- What is the expected stock return that makes you indifferent between stocks and the riskless asset?

Risk aversion

γ	Expected return
0	1%
0.5	2%
1	3%
2	4%
3	6%
4	8%
5	9%
6	11%
7	13%
8	15%
9	16%
10	18%

Indifference curves

- Investors are indifferent between any portfolio that achieves the save expected utility.

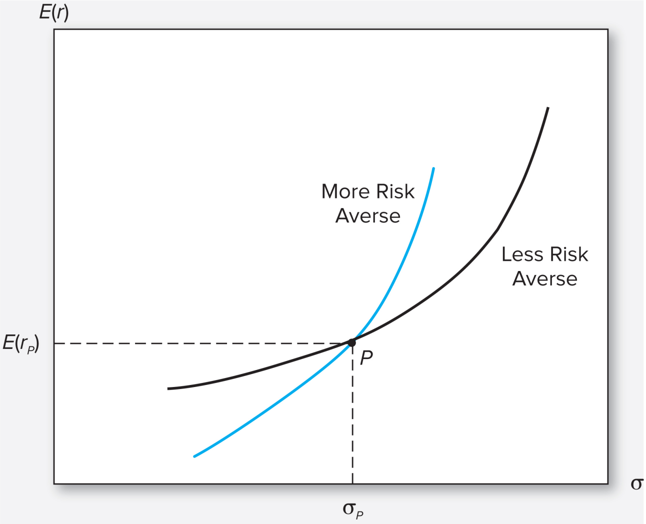
$$\mathbb{E}[U] = \mathbb{E}[R_p] - \frac{\gamma}{2}\mathrm{Var}(R_p)$$

- Rewrite as

$$\mathbb{E}[R_p] = \mathbb{E}[U] + \frac{\gamma}{2}\sigma(R_p)^2$$

- This is a quadratic curve (recall $y = a + bx^2$).
- The intercept is $\mathbb{E}[U]$ is called the certainty equivalent, which is the riskless rate that makes you indifferent.

Indifference curves



Source: Bodie, Kane, and Marcus (2018, Figure 6.2)

Mean-variance portfolio choice

- Portfolio-choice problem:

$$\max_{w} \ \mathbb{E}[U] = \mathbb{E}[R_p] - \frac{\gamma}{2}\mathrm{Var}(R_p)$$

where

$$R_p = R_b + w(R_s - R_b)$$

  - Riskless rate R_b is constant.
  - Stock return R_s has mean μ and variance σ^2.

- Substituting R_p into expected utility,

$$\mathbb{E}[U] = R_b + w(\mu - R_b) - \frac{\gamma}{2}\sigma^2 w^2$$

Solve for the optimal portfolio

- First-order condition:

$$\frac{d\mathbb{E}[U]}{dw} = \mu - R_b - \gamma\sigma^2 w = 0$$

- Second-order condition:

$$\frac{d^2\mathbb{E}[U]}{dw^2} = -\gamma\sigma^2 < 0$$

Optimal portfolio

- Optimal portfolio is

$$w = \frac{\mu - R_b}{\gamma\sigma^2}$$

- Portfolio share in stocks is
  - Increasing in expected excess return $\mu - R_b$.
  - Decreasing in variance σ^2.
  - Decreasing in risk aversion γ.
- Unless infinitely risk averse, always put some money in stocks!

Example 2: Optimal portfolio in June 2022

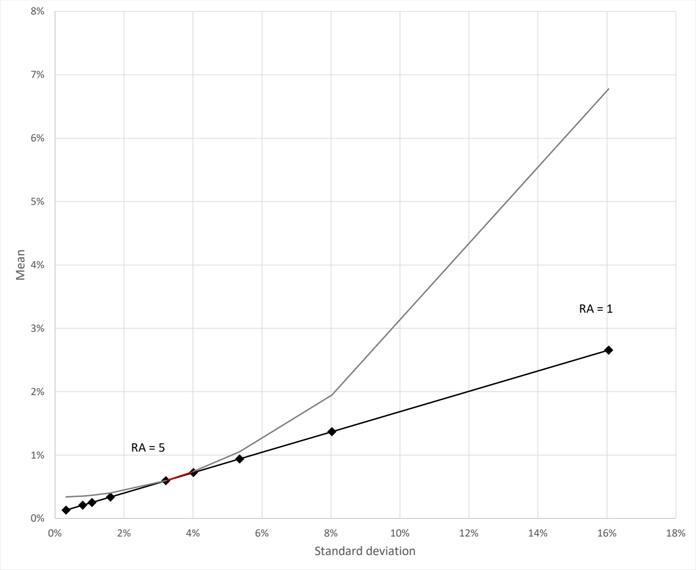
- Plug in the numbers from June 2022.

$$w = \frac{0.94\% - 0.08\%}{\gamma(5.33\%)^2}$$

γ	Weight
1	3.01
2	1.51
3	1.00
4	0.75
5	0.60
10	0.30
15	0.20
20	0.15
50	0.06

- Leverage or buying on margin: $w > 1$ means that the weight on the riskless asset is $1 - w < 0$.

Optimal portfolio



Rebalancing

- Once you optimize your portfolio, do you need to do anything?
- The share of your wealth in stocks after a month is

$$w_t = \frac{w_{t-1}(1 + R_{s,t})}{(1 - w_{t-1})(1 + R_{b,t}) + w_{t-1}(1 + R_{s,t})}$$

  - If stock returns are higher than the riskless rate (i.e., $R_{s,t} > R_{b,t}$), the share of your wealth in stocks increases (i.e., $w_t > w_{t-1}$).
- Assuming that your risk aversion and the mean/variance of stock returns does not change, the theory prescribes a constant w.
- Rebalancing: If stocks returns are higher than the riskless rate, you need to sell stocks and buy the riskless asset.

Example 3: Importance of rebalancing

- Go back to example 1 to illustrate the importance of rebalancing.
- Let's start with $w = 0.60$ in June 1926.
- If you do not rebalance, your wealth in T-bills at date T is

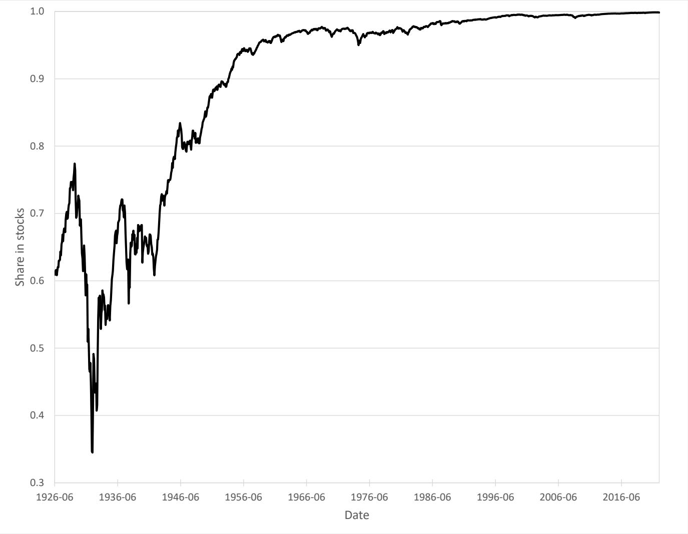
$$B_T = (1 - w) \prod_{t=1926:07}^{T} (1 + R_{b,t})$$

- Your wealth in stocks is

$$S_T = w \prod_{t=1926:07}^{T} (1 + R_{s,t})$$

- At date T, your share in stocks is $S_T/(B_T + S_T)$.

Share in stocks without rebalacing



Importance of rebalancing

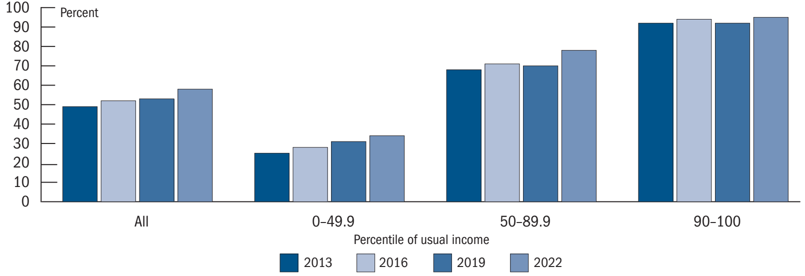
- Without rebalacing, the portfolio share in stocks wanders randomly in the short run and converges to 1 in the long run.
- Rebalancing may be difficult because of behavioral biases.
  - Inertia: We are lazy.
  - Extrapolative expectations: We falsely think that stock returns are going to be high in the future after recent gains.
- Luckily, many accounts now allow you to select an option to automatically rebalance periodically (e.g., once a year on your birthday).

Summary

- Invest a higher share of your wealth in stocks if you are more risk tolerant.
- Unless infinitely risk averse, always put some money in stocks!
- Rebalance your portfolio.

Do US households invest in stocks?

Figure A. Families with direct and indirect holdings of stock, by usual income group, 2013–22 surveys



Note: Key identifies bars in order from left to right.

Source: Survey of Consumer Finances

Next steps

- Should you simply invest in the market portfolio of US stocks?
- Or can you do better by deviating from market weights?
- In particular, should try to “stock pick” (pick only a few stocks to hold)?
- How should you diversify your portfolio internationally?